Kick-off Diagnostic · L3-3 — Identities with Polynomials

Grade 10

8 questions · 30 seconds each · about 4 minutes · Diagnose phase (6 min of 60)

Pre-session anchor (stations 1 and 2 of the pre-session say this, so items 2 and 6 test it).
The one idea: An identity is true for every value of x; $(a+b)^2 = a^2 + 2ab + b^2$ and $(a+b)(a-b) = a^2 - b^2$ let you expand, factor and calculate quickly, and row n of Pascal's triangle gives the coefficients of $(a+b)^n$.
The mistake most students make: Writing $(a+b)^2 = a^2 + b^2$ and losing the middle term $2ab$.
Readiness items 3, 4, 5, 7, 8 test: expanding two brackets, the square and difference-of-squares patterns, and testing a statement with a number.
Wayground set-up: timer 30 s on every question; shuffle options on, questions off; hide the leaderboard; show answers after the quiz, not after each item. Item 1 is a poll (no points). Items 2 to 8 are 1 point each. Tags below: **maroon** = tells you about the pre-session, **navy** = readiness for today. **Green** = correct.

1 POLL · 30 s — SELF-REPORT

Be honest: how much of the pre-session task did you do?

A. All of it, including the readiness check
B. I watched and read but skipped the check
C. I opened it but did not finish
D. I did not open it

Signal. Ungraded. Tally it first; it is the self-report half of the pre-session test.

2 MULTIPLE CHOICE · 30 s — PRE-SESSION CHECK

In the pre-session, the coefficients of $(x+y)^4$ came from row 4 of Pascal's triangle. Which row is it?

✓ **A.** 1, 4, 6, 4, 1
B. 1, 3, 3, 1
C. 1, 5, 10, 10, 5, 1
D. 1, 4, 4, 1

Signal. Wrong → skipped station 1. B = row 3; C = row 5; D = guessed a symmetric row.

3 FILL IN THE BLANK · 30 s — READINESS

Complete the product: $(x+5)(x-5) = x^2 - $ _____

Accept: 25

Signal. 10 = kept a middle term; 5 = did not square the 5. Difference of squares is not yet secure.

4 CHECKBOX · 30 s — READINESS

Select the **two** correct expansions.

✓ **A.** $(x+4)^2 = x^2 + 8x + 16$
✓ **B.** $(x-3)^2 = x^2 - 6x + 9$
C. $(x+2)^2 = x^2 + 4$
D. $(x-5)^2 = x^2 - 25$

Signal. C = lost the middle term $4x$; D = treated a square as a difference of squares.

5 MATCH · 30 s — READINESS

Match each expression with its factored form.

$x^2 - 9$	→	$(x-3)(x+3)$
$x^2 + 6x + 9$	→	$(x+3)^2$ (shown shuffled in Wayground)
$x^2 - 6x + 9$	→	$(x-3)^2$

Signal. Mixing the last two = sign of the middle term; $(x-3)^2$ for $x^2 - 9$ = missed that a difference of squares has no middle term.

6 MULTIPLE CHOICE · 30 s — PRE-SESSION CHECK

The pre-session warned that $(x+3)^2$ is not $x^2 + 9$. Which is the correct expansion?

A. $x^2 + 9$
✓ **B.** $x^2 + 6x + 9$
C. $x^2 + 3x + 9$
D. $x^2 + 6x + 6$

Signal. Wrong → the middle term is the gap. A = squared each term; C = used $x \cdot 3$ once; D = added $3 + 3$ at the end.

7 REORDER · 30 s **READINESS**

Put the steps in order to prove the identity $(x+2)(x-2) = x^2 - 4$.

Correct order: $(x+2)(x-2) \to x^2 - 2x + 2x - 4 \to x^2 - 4 \to$ true for every x, so it is an identity

Signal. Wrong → expanding every term of one bracket over the other is the gap; it is today's proof method.

8 DROPDOWN · 30 s **READINESS**

Is $(x+1)^2 = x^2 + 1$ an identity? Test it at $x = 2$: the two sides are…

✓ **A. 9 and 5, so it is not an identity**

B. 9 and 9, so it is an identity

C. 5 and 5, so it is an identity

D. 3 and 5, so it is not an identity

Signal. B and C = evaluated a side wrongly; D = did not square the bracket. One failing value is enough to disprove an identity.

Read the results (90 seconds)

Readiness R — **Items 3, 4, 5, 7, 8 (out of 5). 0–2** → re-teach expanding two brackets and the three square patterns with me. **3–4** → straight to proving identities and the binomial theorem; fix the missed item in passing. **5** → start the Investigate prompt now.

Pre-session P — **Items 2 and 6 (out of 2), read with the item 1 poll.**

What you see	Reading	What you do
Poll "all" or "read", $P = 2$	Prepared	Nothing extra. Use their R score to route.
Poll "all" or "read", $P = 0$	Claimed, did not retain	Two-minute recap of the one idea at the start. Ask them to restate it in their own words.
Poll "opened" or "did not open"	Task not done	No penalty. Pair with a prepared partner for Targeted Instruction; note it for your task tracker.
$R \leq 2$ and $P = 2$	Content gap	Re-teach group. The task was done, so the idea needs a different explanation.
$R \geq 4$ and $P \leq 1$	Task skipped, skills fine	Move on. Chase the task, not the content.

Class-level rule. If more than 30% of the class chose "opened" or "did not open", or the class average on P is below 1, spend three minutes recapping the one idea above before Targeted Instruction. If more than 30% score $R \leq 2$, the whole class takes the re-teach route first.